

An Emergent Understanding of Mentor Strategies for Career Development in Emerging Fields

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ABSTRACT

Mentoring is a key part of career development, especially in emerging fields such as social entrepreneurship. Internet technologies have made it easier for novice social entrepreneurs to identify and connect with mentors online. Yet, we do not know the strategies mentors use to advise professionals navigating emerging fields. Knowing these strategies would allow us to create technologies for more effective career mentoring. This paper presents an expert model of career mentoring for novice social entrepreneurs. To build the model, we conducted a retrospective cognitive task analysis with 9 mentors who have at least 5 years of experience advising novice social entrepreneurs. We found that mentors help novice social entrepreneurs regulate career-related stress and make decisions about next career steps. Our findings suggest that further exploration of career mentoring strategies might allow designers to develop technologies based on the expert model, such as intelligent agents, to support and scale mentorship in emerging fields.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools**; *User centered design*.

KEYWORDS

mentorship, career development, social entrepreneurship, expert model

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1 INTRODUCTION

Technology is radically changing how people discover new careers by making career pathways more transparent, such as through online communities and networking platforms. Online communities like *Glassdoor* allow people to read employee's opinions about a workplace [11]. Networking platforms such as *LinkedIn*, allow people to see where people work, their formal education, and employment history [36]. People can use information from these online sources to identify and reach out to mentors who can provide advice for effective career development [1, 26]. In more traditional fields, such as entrepreneurship, 84% of professionals rely on mentors to navigate their careers more effectively [12]. Similarly, in academic careers, 73% of students rely on career mentorship from faculty [41].

Career mentoring is especially critical for people to effectively navigate emerging fields, such as social entrepreneurship, because of the lack of career transparency. That is, it is unclear what experiences or skills are necessary to become a social entrepreneur [38]. We focus on social entrepreneurship as an emerging field because social entrepreneurs create wide-scale social and environmental impact as they pursue careers to discover, define, and explore solutions to overcome societal problems [13]. Moreover, navigating social entrepreneurial careers—developing interests, making decisions, and persisting—is difficult, especially for novices [2, 34]. As a result, mentors are especially critical to helping novice social entrepreneurs navigate their careers [30, 41].

Existing HCI research on career mentoring focuses on understanding and designing technologies to help people identify and reach out to mentors [24, 40], but we do not know how to design technology to help mentors provide effective career advice. To understand how technology can help mentors provide effective career advice, it is critical to understand mentorship interactions after the initial connections are made. Yet, we do not know the strategies mentors use to advise professionals navigating emerging fields.

In this study, we conducted 9 retrospective interviews with mentors who have experience advising social entrepreneurs navigating their early careers to answer the following question: *What strategies*

do mentors use to advise novice social entrepreneurs navigating their careers?

We present an emergent expert model of mentorship strategies for career development and discuss how the model might be used for building career mentoring technologies for mentors advising people navigating emerging fields.

2 BACKGROUND

Technology has changed how people find and develop their careers. For example, social media and online communities allow people to connect with peers and mentors [40]; crowdfunding platforms allow people to build financial capital directly from supporters [17]; online job seeking and freelance work sites allow people to find more job opportunities and develop skills required to achieve their desired career goals [5, 15].

However, navigating careers in emerging fields, such as social entrepreneurship, remains difficult in part because the pathways for social entrepreneurship are not as well established as for careers such as law or medicine. This leads to a dearth of public role models. As a result, only those with privileged networks might have access to role models, such as through their family or peer circles [3].

2.1 Social Entrepreneurship: An Emerging Field

Social entrepreneurs play an important role in addressing pressing social and environmental problems, such as climate change, disease, and poverty. Working in areas such as management, design, activism, and education, they may discover, define, and explore solutions to overcome social and environmental problems either in existing or new organizations [44]. For example, a computer scientist interested in climate change might start a new venture for connecting consumers with local produce. Or an architect interested in sustainable urban design might join a nonprofit to work on community development projects. The formal training and mentoring that these professionals receive is not sufficient to navigate this emerging field, because there are challenges such as finding partners, customers, and investors that cross field boundaries and thus require mentors with different expertise to be effectively addressed [16].

Social entrepreneurs are vital to society, but their career paths are not transparent, because fewer people have pursued career paths in this emerging field, leading to a lack of examples for novices to follow. For example, college students interested in pursuing academic careers can use faculty CVs on their websites to understand what a faculty career path might look like. Software engineers might use *LinkedIn* to look up multiple experienced peers so that they might understand what classes to take and skills to develop in internships. In emerging fields like social entrepreneurship, however, fewer people have pursued such a career pathway, making it unclear what experiences or skills are necessary to become a social entrepreneur.

Consequently, novice social entrepreneurs might not be able to understand what a viable career pathway looks like, thus we argue it's more important to provide quality career mentorship for social entrepreneurs.

In prior HCI literature, researchers are generally interested in offering recommendations on how technologies should and should

not be designed to support social entrepreneurship. While some focus on understanding how social entrepreneurs currently interact with existing technology, such as using social media to organize the public to act [29], and using online payment systems to gain funding [21]. Others in the HCI literature aim to understand the context of entrepreneurship to inform technology design for social entrepreneurs. For example, Bidwell [4] studied how rural women managed societal expectations as they contributed to social entrepreneurship efforts, and Crabtree and Chamberlain [9] studied social entrepreneurs' goals for their organizations. This study contributes to the latter literature in understanding the context of social entrepreneurship to inform technology design. Specifically, we look to fill a gap in the HCI literature by studying mentorship during the early career development of social entrepreneurs.

2.2 HCI for Career Development

Previous HCI research has described how technology can support career development processes such as skill development [7], job seeking [14, 15], fundraising [17, 18], and mentoring [24, 40]. Research on technologies for supporting career mentoring are especially important for emerging fields like social entrepreneurship, because their findings may allow practitioners to better support mentors who help people navigate career pathways that are not transparent.

To support career mentoring, researchers in the HCI community have designed and studied systems that help people connect with and seek help from mentors [24, 40]. Building on prior research on help-seeking, Hui et al. [24] found that a web-based tool featuring expert-informed checklist, tagged peer examples, self-tagging, and suggested word limit can help people write more effective introductory help requests to potential mentors. Similarly, Tomprou et al. [40] found that online communities can be rich resources for exchanging career advice through features like Q&A forums.

However, the HCI community has not explored the strategies that mentors use after initial connections are made, which is when mentorship becomes essential for effective career development [27]. To help mentors provide effective career advice, it is critical to understand mentorship interactions after its initial stages.

Prior research defines mentoring and related terms (e.g., coaching) in many ways. We use the term mentor from management science to mean individuals who are named by novice social entrepreneurs as taking an active interest in and providing advice to advance novices' careers [22].

This paper contributes an initial exploration of mentor strategies for helping people navigate emerging fields like social entrepreneurship. We discuss how knowing these strategies might allow HCI researchers to develop technologies for mentors who advise people navigating emerging fields.

3 METHODS

To explore the strategies mentors use to advise novice social entrepreneurs, we conducted retrospective cognitive task analyses (CTA) with mentors reflecting on the last time they interacted with a novice social entrepreneur. This is an appropriate method for

exploring mentoring strategies because it allows us to capture otherwise private interactions that are critical for career development. We analyzed the data following an iterative coding process.

3.1 Participants and Setting

We interviewed 9 mentors based in the United States, such as university instructors (3), graduate students (1), managers at nonprofits (3) and for-profits (1), and experienced social entrepreneurs (1), who have at least 5 years of experience mentoring novice social entrepreneurs navigating their careers. Mentors were screened through public online information and interview questions.

To recruit these mentors, we used snowball sampling [20] from novice social entrepreneurs who identified mentors who helped them navigate their careers. That is, we first asked novice social entrepreneurs over email to identify mentors who have helped them in their career. We identified novice social entrepreneurs through the social networks of the authors and of other research lab members. Snowball sampling is appropriate in this case, because it allows us to identify a difficult to access population, mentors in novice social entrepreneurs' networks who help them to navigate their early careers. To address the bias issue with snowball sampling, we contacted a diverse sample of novice social entrepreneurs across different fields (e.g., education, design, sustainability), locations (e.g., universities, states) and relationships with members of the research team (e.g., friends, family members, colleagues, students).

We operationalized novice social entrepreneurs as recent college graduates (1-2 years out of college) who are either working at nonprofits or starting their own social enterprises. This is an appropriate operationalization of novice social entrepreneurs because prior research has identified these two career pathways as antecedents to pursuing a full-time social entrepreneurial career [35]. We focus on recent college graduates because prior literature identifies this transition as a critical period in career development [32].

3.2 Data Collection

To build an expert model of mentoring strategies, we conducted retrospective cognitive task analyses (CTA) [39] following the critical decision method [43] in interviews with mentors. To help the mentor recall a specific recent mentoring interaction, we asked participants to reflect on the last time they interacted with a novice social entrepreneur about their mentee's careers. They were instructed to think aloud about what advice they provided and how they decided to provide such advice. This is an appropriate interviewing method because it allows researchers to understand the underlying cognitive mechanisms that inform decisions made by expert mentors, which can then be translated into mentoring support and training technologies (e.g., [33]). As a result, the purpose of these interviews was to understand the strategies mentors use to advise novice social entrepreneurs instead of understanding the role of technology in these interactions. That is, we did not ask mentors about how technology influences their mentoring process. These interviews lasted for 30-45 minutes and were transcribed for analysis.

3.3 Analysis and Expert Model

We analyzed the retrospective CTA data using the emergent themes approach [43] following an iterative coding process [31] because the purpose of this analysis was to create an empirically-grounded expert model of mentoring strategies. Using the transcribed data, we identified and categorized mentor strategies. We began with rounds of inductive coding and identified codes such as *state no perfect job*, *reduce stress*, and *offer criteria*. Then, using the method of constant comparisons [8, 31], we merged codes into preliminary themes of closely related concepts, such as *help make decisions* and *help manage stress*. Finally, we iteratively reviewed and discussed preliminary themes to arrive at final themes as shown in Figure 1.

To represent the expert model, we organize mentors' cognitive strategies following Smirnov and colleagues' [37] example of hierarchical task analysis [10]. This is an appropriate form of knowledge representation for this project because it includes perceptions, action, and goals structures that led mentors to use their strategies.

4 FINDINGS

In this section, we present an emergent expert model of how mentors advise novice social entrepreneurs navigating their early careers. We found that mentors help novice social entrepreneurs regulate career-related emotions and make decisions. Although not the focus of this study, some mentors also describe the role of technology in their interactions with novices.

4.1 Mentors Help Novice Social Entrepreneurs Regulate Career-Related Stress

We found that mentors perceive novice social entrepreneurs to be often feeling stressed due to high expectations caused by social comparisons, family, and job seeking (5 out of 9). Mentors try to address their stress by stating that there's no perfect job (4 out of 9). For example, an university instructor identify reasons for mentees to be feeling stressed during their undergraduate degree:

There's always a lot of stress around like "What should I do over the summer?" and I'm like, "It doesn't matter, just pick something. And if you want to, make your own opportunities to do that work wherever you are, whatever you're doing." [...] The culture is very rigid about where students expect that they should go. Fifty percent of our undergrads are pre-med, so those students have a pretty linear path in mind. And if you don't know where you're going, you really stick out. A lot of times it's anxiety or frustration and they're comparing themselves to their peers and feeling like they don't have a path. Or the anxiety is feeling like their parents think that they need to get something marketable and so they need to go get the business of social innovation minor and they're freaked out about finishing it. (P2, university instructor)

In this quote, the mentor reflects on how they perceive that their mentees experience considerable stress in navigating a social entrepreneurial career, which they perceive being a non-linear career path, as opposed to a linear path like medicine. They suggest sources of stress related to social comparisons, family expectations,

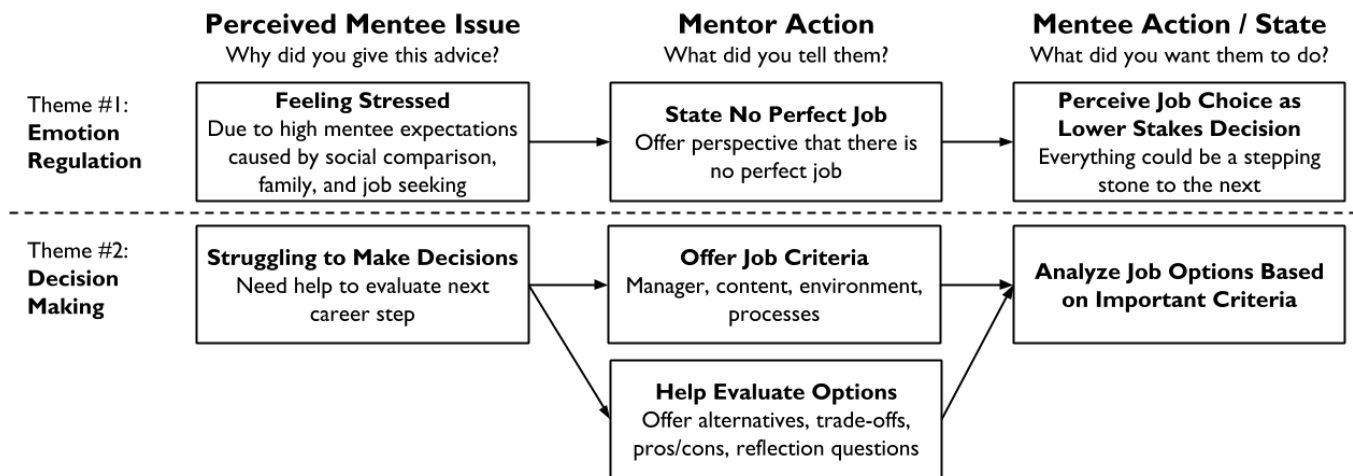


Figure 1: Emerging expert model of mentor strategies for advising novice social entrepreneurs.

and job seeking that mentees balance when making a career decision, which suggests that novice social entrepreneurs experience career-related stress from multiple sources.

To help novice social entrepreneurs reduce the stress associated with navigating their careers, mentors state that there is no perfect job. For instance, the manager at a nonprofit stated to their mentee:

I especially felt this when I was your age. That you feel like the next job that you get has to be like the perfect one because it's going to be like the key to your future. It's gonna unlock this pathway that you're supposed to go down, or all these things. In reality, what I've learned is that there is not a perfect job, but everything could be a perfect stepping stone to your next. (P4, nonprofit manager)

Here, the mentor offered their own experience feeling stressed about choosing jobs and then stated that there is no perfect job. This quote suggests that mentors hope that, by stating that there is no perfect job, mentees will perceive lower stakes when choosing a job. This quote also suggests that although the Internet has made it easier for job seekers to browse opportunities, those navigating emerging fields might feel overwhelmed by choice, especially when the majority of roles available are in more traditional fields.

4.2 Mentors Help Novice Social Entrepreneurs Make Decisions About Next Career Step

We also found that, from the mentor's perspective, novice social entrepreneurs struggle to make decisions about their next career step (5 out of 9). Mentors try to help them by offering criteria for them to consider when evaluating their options (4 out of 9) or by directly helping them to evaluate possible options (3 out of 9). For example, an university instructor talks about how a mentee used a networking platform to contact them because they were struggling to decide a major in college:

She just sent me a message on LinkedIn and was like "Hey, I heard about you and like I'm confused about what

I'm doing, so can we talk?" We had like a 30 minute phone call, which was pretty introductory. She was like "I don't think engineering is the right thing for me, but I don't know what to do instead." (P2, university instructor)

In this interaction, the mentor describes how a mentee found and contacted them over an online networking platform to seek advice about choosing a field of study. Similarly to other interactions (P7), the mentor is helping their mentee decide between fields (e.g., engineering or design). Mentors also advise novice social entrepreneurs deciding between jobs, roles, or positions (P3, P5, P6).

To help novice social entrepreneurs make decisions about their next career step, mentors offer criteria that their mentees should consider. For example, another nonprofit manager offers a framework for their mentees to use when making decisions about jobs and fields:

There's a couple different levels to think about careers, so I always start at the highest level: What is the problem you want to solve? What is the thing in the world that makes you uncomfortable? The next level is organizational, environmental so: How big is the organization that you want to work on? What's its component or solution proposed to the problem? I think the UN could be working on the same problem that a small social enterprise could be working on. And then the last level is: What do you want to do every day? How are you actually applying your skills to the mission? I think for me, I really started thinking about work that way when it took me forever to find a job. (P9, nonprofit manager)

This mentor is helping novice social entrepreneurs make a decision by considering criteria such as the content ("the problem you want to solve"), the environment (size of the company and proposed solution), and the processes ("what do you want to do every day") of each job through guiding questions.

Mentors also help their mentees make decisions by directly helping them evaluate options by raising trade-offs between the options they are considering:

I didn't want to push him to one decision or another, but more so just tried to ask him some questions like, "What's your purpose for going [to graduate school]?" and then think through some of the trade-offs so that when [mentee] tried to make a decision, it could be a little bit more informed. (P5, graduate student)

Here, the mentor helps the mentee make a more informed decision by talking through potential trade-offs that they would have to make if they decided to pursue graduate school as their next career step. These quotes suggest that mentors hope that mentees will make a more informed decision by evaluating their options using the criteria proposed and trade-offs raised by the mentor.

5 DISCUSSION

Despite the increase in technologies for supporting career development through mentorship, there is still a lack of understanding about the strategies mentors use to advise professionals, especially in emerging fields, where mentorship is especially important. While exploring these strategies, we found that mentors help novice social entrepreneurs regulate career-related stress and make decisions about their next career step. These findings extend the existing mentoring framework developed by Higgins and Kram [22] by exploring the perceptions and goals that mentors have when advising novice social entrepreneurs. We discuss design implications for supporting and scaling mentorship as well as limitations of this work.

5.1 Design Implications

We created a basic expert model of what strategies mentors use to advise novice social entrepreneurs navigating their careers. With this model, future work can explore how we might design intelligent agents for helping mentors provide more effective advice. For example, similarly to online counseling systems (e.g., [19, 28]), future work can focus on designing wizards to help mentors diagnose common mentee issues, such as feeling stressed by comparing themselves with peers or struggling to make a complex career decision.

This expert model will also allow designers to create intelligent tutoring systems that provide automated mentoring (e.g., [6, 42]). For example, online job seeking platforms like *Indeed* could offer a chatbot that provides expert-informed advice to job seekers based on their job search. Automated mentoring might be promising for professionals who might not have access to a wide network of mentors.

Another potential direction would be to design training systems or mixed-initiative systems [23] that make it easier for mentors to provide advice. For example, career advisors or online mentors [25] could use an expert-informed, online checklist to advise students navigating emerging careers during 1-on-1 advising sessions.

5.2 Limitations and Future Work

This initial study is not without limitations. While this preliminary study included data from 9 mentors of novice social entrepreneurs,

future work will address this issue by contacting and interviewing more mentors until we reach data saturation. Moreover, we explore the strategies mentors use to advise professionals in only one emerging field: social entrepreneurship. In order to generalize these findings to other emerging fields, such as human-centered design or civic technology, future work should identify and interview mentors in different areas. We will also validate our model of mentorship strategies with mentors. Another limitation of this initial study is that we do not know if the mentoring strategies mentors use with novice social entrepreneurs are similar to those navigating other career paths. As a result, some of the strategies presented on this paper might also apply to professionals navigating traditional fields. We focus on social entrepreneurship because mentorship is especially important for their career development. Future studies should explore mentoring strategies in different fields.

6 CONCLUSION

Existing HCI research on career mentoring focuses on understanding and designing technologies to help people identify and reach out to mentors. However, we have not explored how to design technologies to help mentors provide effective career advice. To address this gap, we contribute an expert model of the strategies mentors use to help social entrepreneurs navigate their careers to the fields of HCI, social entrepreneurship, and career development. We found that mentors perceive novice social entrepreneurs to feel stressed from high expectations created from social comparisons, family expectations, and job seeking. One strategy that mentors use to reduce career-related stress is to state that there is no perfect job. Mentors also help novice social entrepreneurs make career decisions. They do that by offering criteria and by raising trade-offs for their mentees to consider. Our findings extend existing work on HCI systems for career development by exploring career mentoring strategies in the context of social entrepreneurship. These findings help us build better career support systems for those navigating other emerging fields and likely traditional fields as well. With greater support for implementing strategies for mentoring novice professionals navigating emerging fields, people can more effectively leverage the connections Internet technologies can offer.

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